

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("/content/train.csv - train.csv.csv")
```

```
print(df.head())

print("\nDataset Shape:")
print(df.shape)

print("\nDataset Information:")
print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())

print("\nStatistical Summary:")
print(df.describe())
```

	PassengerId	Survived	Pclass	\
0	1	0	3	
1	2	1	1	
2	3	1	3	
3	4	1	1	
4	5	0	3	

	Name	Sex	Age	SibSp	\
0	Braund, Mr. Owen Harris	male	22.0	1	
1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	
2	Heikkinen, Miss. Laina	female	26.0	0	
3	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	
4	Allen, Mr. William Henry	male	35.0	0	

	Parch	Ticket	Fare	Cabin	Embarked
0	0	A/5 21171	7.2500	NaN	S
1	0	PC 17599	71.2833	C85	C
2	0	STON/O2. 3101282	7.9250	NaN	S
3	0	113803	53.1000	C123	S
4	0	373450	8.0500	NaN	S

Dataset Shape:
(891, 12)

Dataset Information:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  -
0   PassengerId  891 non-null   int64
1   Survived     891 non-null   int64
2   Pclass       891 non-null   int64
3   Name         891 non-null   object
4   Sex          891 non-null   object
5   Age          714 non-null   float64
6   SibSp        891 non-null   int64
7   Parch        891 non-null   int64
8   Ticket       891 non-null   object
9   Fare         891 non-null   float64
10  Cabin        204 non-null   object
11  Embarked     889 non-null   object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
None
```

Missing Values:

PassengerId	0
Survived	0
Pclass	0
Name	0

```
Sex      0
Age      177
SibSp    0
Parch    0
Ticket   0
Fare     0
Cabin    687
```

```
df['Age'].fillna(df['Age'].median(), inplace=True)
```

/tmp/ipykernel_809/1933487976.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate ob

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inpla

```
df['Age'].fillna(df['Age'].median(), inplace=True)
```

```
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

/tmp/ipykernel_809/411043493.py:1: FutureWarning: A value is trying to be set on a copy of a DataFrame. The behavior will change in pandas 3.0. This inplace method will never work because the intermediate ob

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inpla

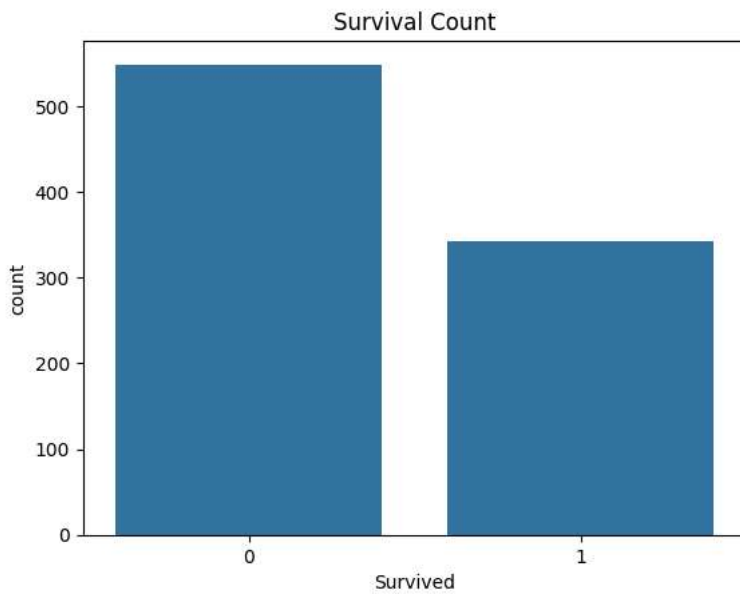
```
df['Embarked'].fillna(df['Embarked'].mode()[0], inplace=True)
```

```
df.drop('Cabin', axis=1, inplace=True)
```

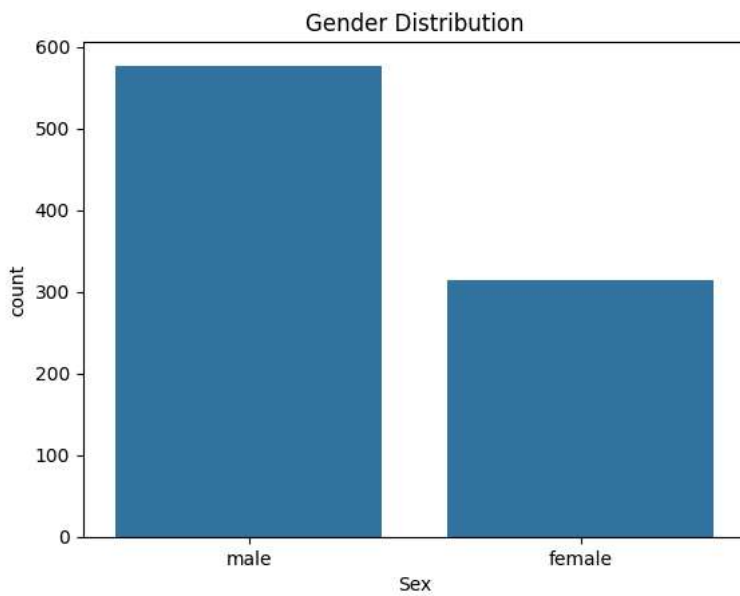
```
print(df.isnull().sum())
```

```
PassengerId    0
Survived        0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64
```

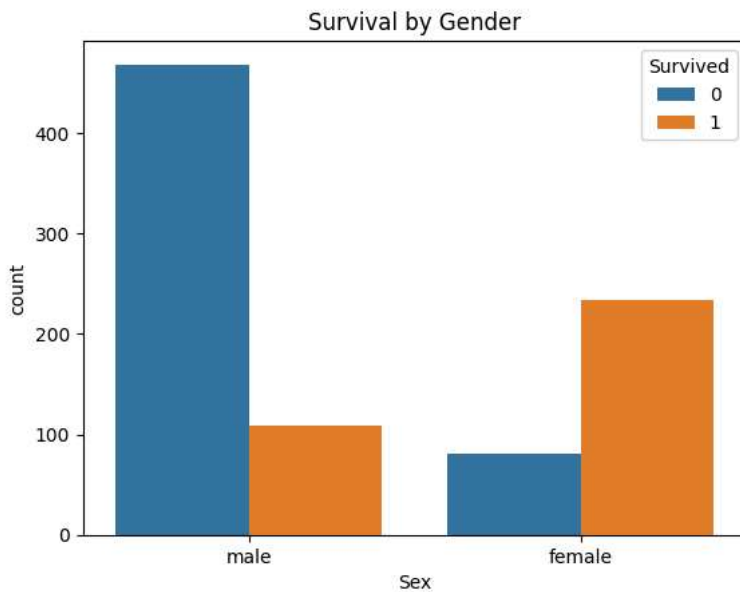
```
sns.countplot(x='Survived', data=df)
plt.title("Survival Count")
plt.show()
```



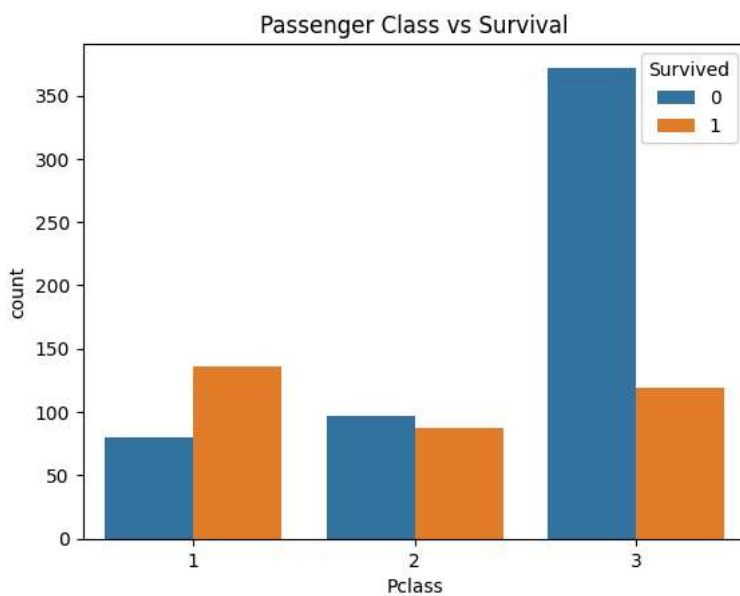
```
sns.countplot(x='Sex', data=df)  
plt.title("Gender Distribution")  
plt.show()
```



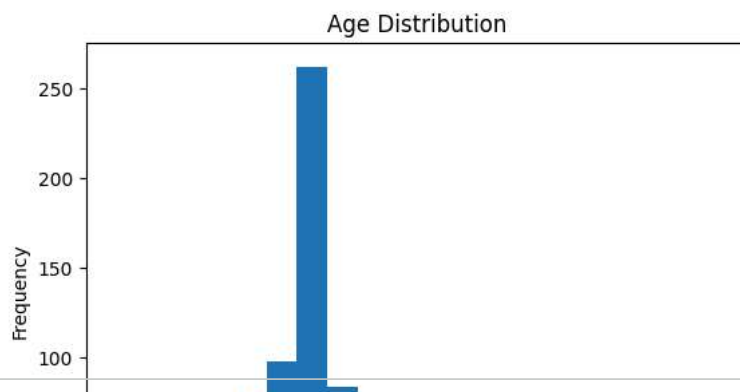
```
sns.countplot(x='Sex', hue='Survived', data=df)  
plt.title("Survival by Gender")  
plt.show()
```



```
sns.countplot(x='Pclass', hue='Survived', data=df)  
plt.title("Passenger Class vs Survival")  
plt.show()
```



```
plt.hist(df['Age'], bins=20)  
plt.title("Age Distribution")  
plt.xlabel("Age")  
plt.ylabel("Frequency")  
plt.show()
```



```
plt.hist(df['Fare'], bins=20)  
plt.title("Fare Distribution")  
plt.xlabel("Fare")  
plt.ylabel("Frequency")  
plt.show()
```

